

Technology #168

Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continuously releases a heparin-mimetic, non-binding surface to inhibit biofilm and heparin-binding thrombus on the catheter lumen

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continuously releases a heparin-mimetic, non-binding surface to inhibit biofilm and heparin-binding thrombus on the catheter lumen

WHY DOES IT MATTER?

Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000003: Other [CARMEDA CIRCUIT - HEPARIN COATED]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'An anticoagulant activity and protein interaction study of chondroitin sulfate-coated magnetic nanocomposites.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 11 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in phage_microbio. 7-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct an in vitro flow loop study comparing coated versus uncoated tunneled catheters, measuring thrombus mass and biofilm colony formi.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: phage infectivity retention vs time on surface. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use; surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific; surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation); BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.; BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .; BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.; Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), $N_{bacteria}$ (-), N_{phage} (-), $N_{bacteria}$ (-) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: N (-), $dN_{infected}/dt$ (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_{phage} , -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r , 1/s] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Biofilm formation on catheter surfaces leading to protein adhesion and thrombus

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'heparin-binding thrombus formation' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters develop heparin-binding thrombus despite antithrombotic luminal coatings?". [fda_maude] 100 records retrieved live (1992-2070). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 1000003: Other [CARMEDA CIRCUIT - HEPARIN COATED]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'Antithrombotic Polymers: A Narrative Review on Current and Future Strategies for Their Design, Synthesis, and Application.'; 'An anticoagulant activity and protein interaction study of chondroitin sulfate-coated magnetic nanocomposites.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: phage_microbio

Disciplines: microbiology, surface engineering, pharmacodynamics, materials science

Architecture: PROPOSED architecture (no physical system exists): phage loading / immobilization layer; release or contact-kill mechanism; surface substrate (coating architecture); stability preservation element — realizing: Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continuously releases a heparin-mimetic, non-binding surface to inhibit biofilm and heparin-binding thrombus on the catheter lumen

Subsystems:

?: phage loading / immobilization layer - ?

?: release or contact-kill mechanism - ?

?: surface substrate (coating architecture) - ?

?: stability preservation element - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continuously releases a heparin-mimetic, non-binding surface to inhibit biofilm and heparin-binding thrombus on the catheter lumen

Key Physics: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Microbiology / phage anti-biofilm governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

BIO-002: Exponential population dynamics. Expression: $N = N_0 * \exp(r * t)$. Applicability to this invention: CONDITIONAL; holds under: unlimited substrate; constant net rate over the window. Assumptions: kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use. Engaged variables: NOT ESTABLISHED.

BIO-003: Surface-limited adsorption kinetics. Expression: $dN_{\text{infected}}/dt = k_{\text{ads}} * N_{\text{phage}} * N_{\text{bacteria}}$. Applicability to this invention: APPLICABLE; holds under: well-mixed suspension; constant k_{ads} . Assumptions: surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation); k_{ads} is strain- and environment-specific. Engaged variables: k_{ads} .

BIO-001: Multiplicity of infection. Expression: $MOI = N_{\text{phage}} / N_{\text{bacteria}}$. Applicability to this invention: CONDITIONAL; holds under: well-mixed system; planktonic or defined-surface assay. Assumptions: surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation). Engaged variables: NOT ESTABLISHED.

Assumptions:

- kill kinetics on an immobilized surface are not well-mixed growth — re-measure before quantitative use
- surface-immobilized phage dosing is not well-mixed — use contact-area density instead (MODEL_DERIVED adaptation)
- k_{ads} is strain- and environment-specific
- surface-immobilized phage dosing is not a well-mixed MOI — use contact-area density instead (MODEL_DERIVED adaptation)
- BIO-002 (Exponential population dynamics) is applicable only under: unlimited substrate; constant net rate over the window.
- BIO-003 (Surface-limited adsorption kinetics) is applicable only under: well-mixed suspension; constant k_{ads} .
- BIO-001 (Multiplicity of infection) is applicable only under: well-mixed system; planktonic or defined-surface assay.

- Model input variables requiring sourced values before any numeric result: N_0 (-), r (1/s), t (s), k_{ads} (m^3/s), N_{phage} (-), $N_{bacteria}$ (-), N_{phage} (-), $N_{bacteria}$ (-) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: N (-), $dN_{infected}/dt$ (1/s), MOI (-) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): surface phage loading density [N_{phage} , -] value_status=UNKNOWN; infectivity retention half-life [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; host range breadth [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; moi at contact surface [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; CFU challenge load [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; resistance emergence rate [r , 1/s] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'heparin-binding thrombus formation', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'infectivity loss before clinical use window', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'resistant strain emergence negating kill', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'host-range mismatch to clinical isolate', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coating wear releasing loaded phage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'biofilm matrix blocking phage access', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	must ensure long-term patency with biocompatible antithrombotic surface	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Biofilm formation on catheter surfaces leading to protein adhesion and thrombosis	MODELLED
DI-003	Expected effect target	Decreased biofilm accumulation and reduced heparin-binding protein adsorption	MODELLED
DI-004	Domain design input pattern: target organ application: biofilm eradication [biofilm]	evergreen applicable biofilm eradication [biofilm]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: device surface applicability: device [surface]	evergreen applicable device [surface]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: stability requirement: NO SUPPLY ABANDONED (candidate pattern; confirm against engineering profile)	evergreen applicable NO SUPPLY ABANDONED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance with the standards and rules (ISO 22196 (antimicrobial surface hygiene))	compliance with the standards and rules (antimicrobial surface hygiene)	ENGINEERING_PROPOSED
DI-008	Custodied observation: Pathology of catheter-related complications (e.g., infection, thrombosis, and inflammation)	https://pubmed.ncbi.nlm.nih.gov/36269766/ catheter-related complications (e.g., infection, thrombosis, and inflammation)	ENGINEERING_PROPOSED
DI-009	Custodied observation: Hallmarks of catheter-related complications (e.g., thrombosis, infection, and inflammation)	https://pubmed.ncbi.nlm.nih.gov/36269766/ catheter-related complications (e.g., thrombosis, infection, and inflammation)	ENGINEERING_PROPOSED
DI-010	Custodied observation: Over-the-wire catheter-related complications (e.g., thrombosis, infection, and inflammation)	https://pubmed.ncbi.nlm.nih.gov/36269766/ catheter-related complications (e.g., thrombosis, infection, and inflammation)	ENGINEERING_PROPOSED
DI-011	Custodied observation: Comprehensive catheter-related complications (e.g., thrombosis, infection, and inflammation)	https://pubmed.ncbi.nlm.nih.gov/36269766/ catheter-related complications (e.g., thrombosis, infection, and inflammation)	ENGINEERING_PROPOSED

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	phage loading / immobilization layer realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that
DO-002	release or contact-kill mechanism realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that con	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that con
DO-003	surface substrate (coating architecture) realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) th	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) th
DO-004	stability preservation element realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-005	phage loading / immobilization layer interfaces with release or contact-kill mechanism realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-006	release or contact-kill mechanism interfaces with surface substrate (coating architecture) realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-007	surface substrate (coating architecture) interfaces with stability preservation element realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-008	Device boundaries: patient/fluid inlet and outlet termination of control/data paths required by application self-definition	UNKNOWN	control/data paths required by application self-definition
DO-009	Flow/force/signal path geometry realizing the mechanism of self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-010	Operating range for surface phage loading density within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-011	Operating range for infectivity retention half-life within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-012	Operating range for host range breadth within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-013	Operating range for moi at contact surface within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-014	Operating range for CFU challenge load within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-015	Operating range for resistance emergence rate within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the mechanism of self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-017	Information path: sensing element -> decision/monitoring element realizing Apply a self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu	UNKNOWN	self-polishing copolymer anti-fouling coating (adapted from marine anti-fouling technology) that continu
DO-018	Fixture enabling: phage infectivity retention vs time on surface within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-019	Fixture enabling: biofilm prevention CFU challenge assay within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-020	Fixture enabling: host-range and resistance-emergence challenge assay within which the mechanism holds its stated behavior	UNKNOWN	stated behavior
DO-021	Fixture enabling: CFU biofilm-prevention challenge assay within which the mechanism holds its stated behavior	UNKNOWN	stated behavior

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
surface phage loading density	UNKNOWN (no sourced value)	-	ENGINEERING_PROPOSED	(design assigned and target value and verification method)
infectivity retention half-life	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target value and verification method)
host range breadth	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target value and verification method)
moi at contact surface	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target value and verification method)
CFU challenge load	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned and target value and verification method)
resistance emergence rate	UNKNOWN (no sourced value)	1/s	ENGINEERING_PROPOSED	(design assigned and target value and verification method)

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
heparin-binding thrombolytic failure	heparin-binding thrombolytic failure	design for heparin-binding thrombolytic failure	design for heparin-binding thrombolytic failure	design for heparin-binding thrombolytic failure
infectivity loss before stabilization	infectivity loss before stabilization	infectivity loss before stabilization	infectivity loss before stabilization	infectivity loss before stabilization
resistant strain emergence	resistant strain emergence	resistant strain emergence	resistant strain emergence	resistant strain emergence
host-range mismatch	host-range mismatch	host-range mismatch	host-range mismatch	host-range mismatch
coating wear releasing phage	coating wear releasing phage	coating wear releasing phage	coating wear releasing phage	coating wear releasing phage
biofilm matrix blocking phage access	biofilm matrix blocking phage access	biofilm matrix blocking phage access	biofilm matrix blocking phage access	biofilm matrix blocking phage access

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
heparin-binding thrombus formation	Failure of heparin-binding thrombus formation	design the device to be established	design the device to be established
infectivity loss before clinical use	Failure of infectivity loss before clinical use	stabilization of phage particles	stabilization of phage particles
resistant strain emergence	Failure of resistant strain emergence	negative phage selection	negative phage selection
host-range mismatch to clinical use	Failure of host-range mismatch to clinical use	isolation of phage's specificity	isolation of phage's specificity
coating wear releasing loaded phage	Failure of coating wear releasing loaded phage	phage-resistant coating	phage-resistant coating
biofilm matrix blocking phage access	Failure of biofilm matrix blocking phage access	degradation of biofilm matrix	degradation of biofilm matrix

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct an in vitro flow loop study	Conduct an in vitro flow loop study	Conduct an in vitro flow loop study	Conduct an in vitro flow loop study
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	phage infectivity retention vs time on surface	phage infectivity retention vs time on surface	phage infectivity retention vs time on surface	phage infectivity retention vs time on surface
VF-004	biofilm prevention CFU challenge assay	biofilm prevention CFU challenge assay	biofilm prevention CFU challenge assay	biofilm prevention CFU challenge assay
VF-005	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm	CFU challenge assay against established biofilm

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	electrospun + metal-coated catheter	semi-proprietary device coatings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
electrospun + metal-coated catheter	domain candidate material	1	Material	electrospun + metal-coated catheter	semi-proprietary device coatings	True	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- electrospinning + immobilization dip under aseptic control: domain registry manufacturing route - candidate only; no process qualification exists
- lyophilization/storage stabilization: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-001', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-002', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-004', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter surface phage loading density [N_phage -]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter infectivity retention half-life [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter host range breadth [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter moi at contact surface [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter CFU challenge load [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}

- {'unknown': 'value of critical parameter resistance emergence rate [r 1/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	conducting the proposed physical verification	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (constraint compliance)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Conduct an in vitro bioassay	NOT ESTABLISHED (VF-001 verification)	NOT ESTABLISHED (VF-001 verification)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment	NOT ESTABLISHED (VF-002 verification)	NOT ESTABLISHED (VF-002 verification)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	phage infectivity retention assay	NOT ESTABLISHED (VF-003 verification)	NOT ESTABLISHED (VF-003 verification)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	biofilm prevention CFU challenge assay	NOT ESTABLISHED (VF-004 verification)	NOT ESTABLISHED (VF-004 verification)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	CFU challenge assay	NOT ESTABLISHED (VF-005 verification)	NOT ESTABLISHED (VF-005 verification)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.